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Design and Optimization of a Leaf Inspired High Efficiency Solar Still

EEB1411-PROJECT PHASE 1

FINAL REVIEW

BATCH - 1

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ABSTRACT

This project presents the Design and Optimization of a Leaf-Inspired High-Efficiency Solar Still for improved freshwater production. The proposed system uses leaf-like branching channels to distribute water uniformly as a thin film over the absorber surface. A black absorber coating, PCM-based thermal storage, and heat-recovery preheating are incorporated to improve thermal energy utilization. The system will be analyzed using MATLAB simulation to study temperature distribution, heat transfer, evaporation behaviour, and freshwater productivity. The design will be optimized by comparing different configurations with a conventional solar still.

SDG GOAL MAPPING

SDG 7 - Affordable and Clean Energy

Utilizes renewable solar energy Reduces dependence on conventional energy sources. Uses PCM for thermal energy storage. Improves solar heat utilization. Supports sustainable and energy-efficient desalination.

SDG 9 - Industry, Innovation and Infrastructure

The project introduces an innovative leaf-inspired solar still design with branching water channels, PCM-based thermal storage, and feedwater preheating. Simulation-based performance analysis supports the development of improved and sustainable solar desalination technology

INTRODUCTION

Solar desalination is a sustainable method of producing freshwater using solar energy. Conventional solar stills operate by heating impure or saline water, evaporating it, and condensing the vapour on a transparent cover. However, conventional designs generally have limited evaporation area, non-uniform water distribution and considerable thermal losses, resulting in lower freshwater productivity. This project proposes a leaf-inspired solar still that uses a biomimetic branching channel network to distribute water as a thin film over the absorber surface. A black solar-absorbing basin is used to enhance heat absorption, while heat recovery and feed-water preheating are proposed to improve thermal utilization. Phase Change Material (PCM) is incorporated for thermal energy storage. The proposed design will be numerically investigated and optimized to improve thermal performance and freshwater production.

PROBLEM STATEMENT

- Conventional solar stills generally use a flat basin with stagnant or relatively deep water, which limits the rate of evaporation.
- Water distribution over the basin may be non-uniform, reducing the effective evaporation area.
- A significant amount of absorbed thermal energy can be lost through the bottom and side walls of the system.
- Incoming feed water enters at a relatively low temperature and requires additional solar energy for heating.
- Thermal energy available during high solar radiation is not effectively stored for periods of reduced solar intensity.
- Conventional systems generally have single-stage evaporation and limited heat recovery.

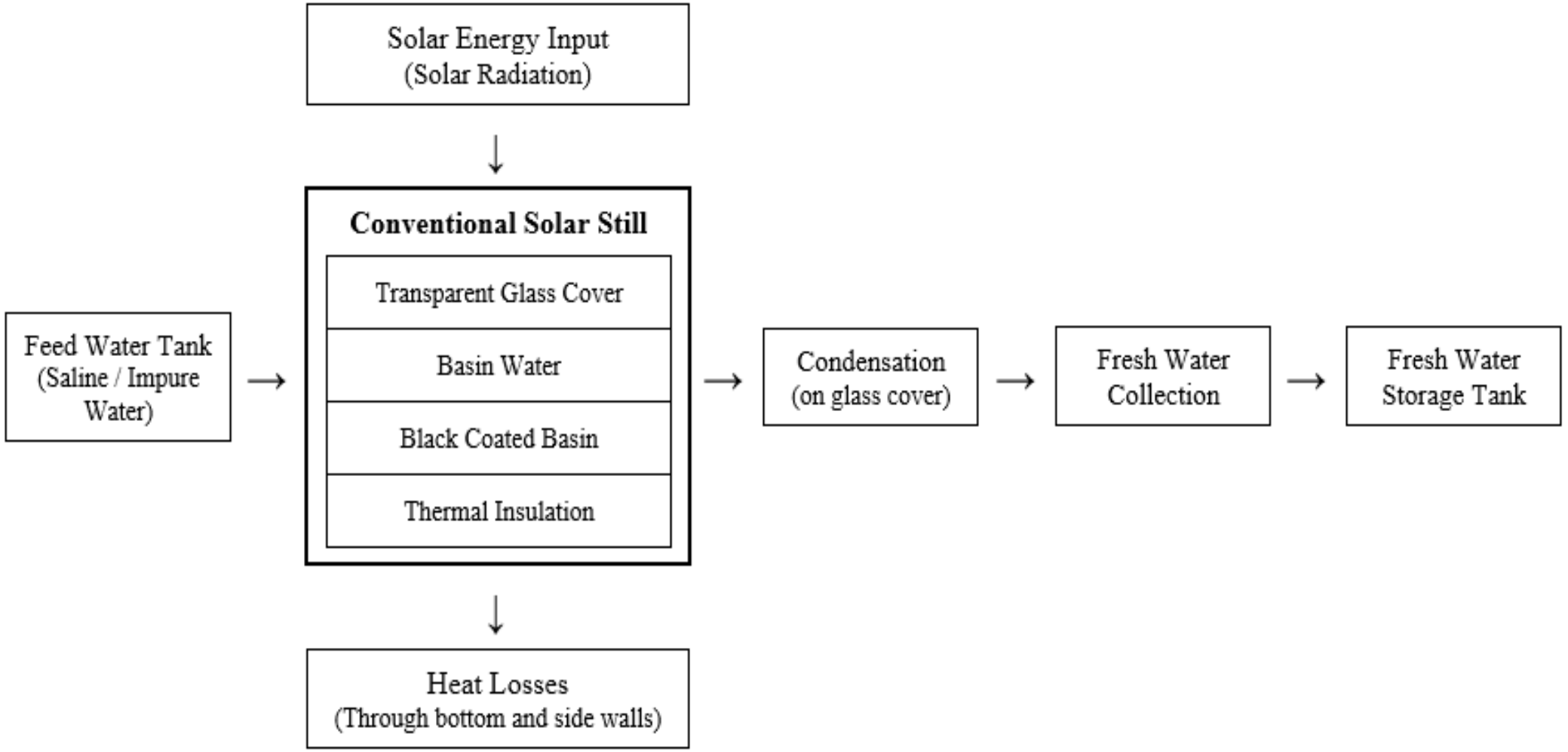
OBJECTIVE

- To develop a leaf-inspired basin geometry with main and branching water channels.
- To introduce thin-film water flow for improved water distribution and evaporation.
- To apply a black solar-absorbing coating to enhance solar heat absorption.
- To investigate feed-water preheating using heat recovery from the hot outlet/condensation side.
- To integrate PCM thermal energy storage for better utilization of available solar heat.
- To compare the proposed design with a conventional solar still under identical operating conditions.

LITERATURE REVIEW

S.NO	TITLE OF THE PAPER	YEAR	AUTHOR	PROBLEMS IDENTIFIED
1	Solar Stills: A Comprehensive Review of Designs, Performance and Material Advances	2016	D. Dsilva Winfred S. Iniyan L. Suganthi P. A. Davies	Conventional solar stills have low freshwater productivity and significant heat loss.
2	Phase Change Materials (PCMs) for Improving Solar Still Productivity: A Review	2020	Hussein A. Mohammed Mehdi Khiadani	Solar still performance is affected by limited thermal energy storage, and PCM effectiveness varies with operating conditions and time of operation.
3	Improved Solar Still Productivity Using PCM and Nano-PCM Composites Integrated Energy Storage	2024	G. Murali, P. Ramani, M. Murugan, P. V. Elumalai.	Conventional systems have limited energy storage and productivity; the study investigates PCM/NPCM integration to improve thermal storage and freshwater production.

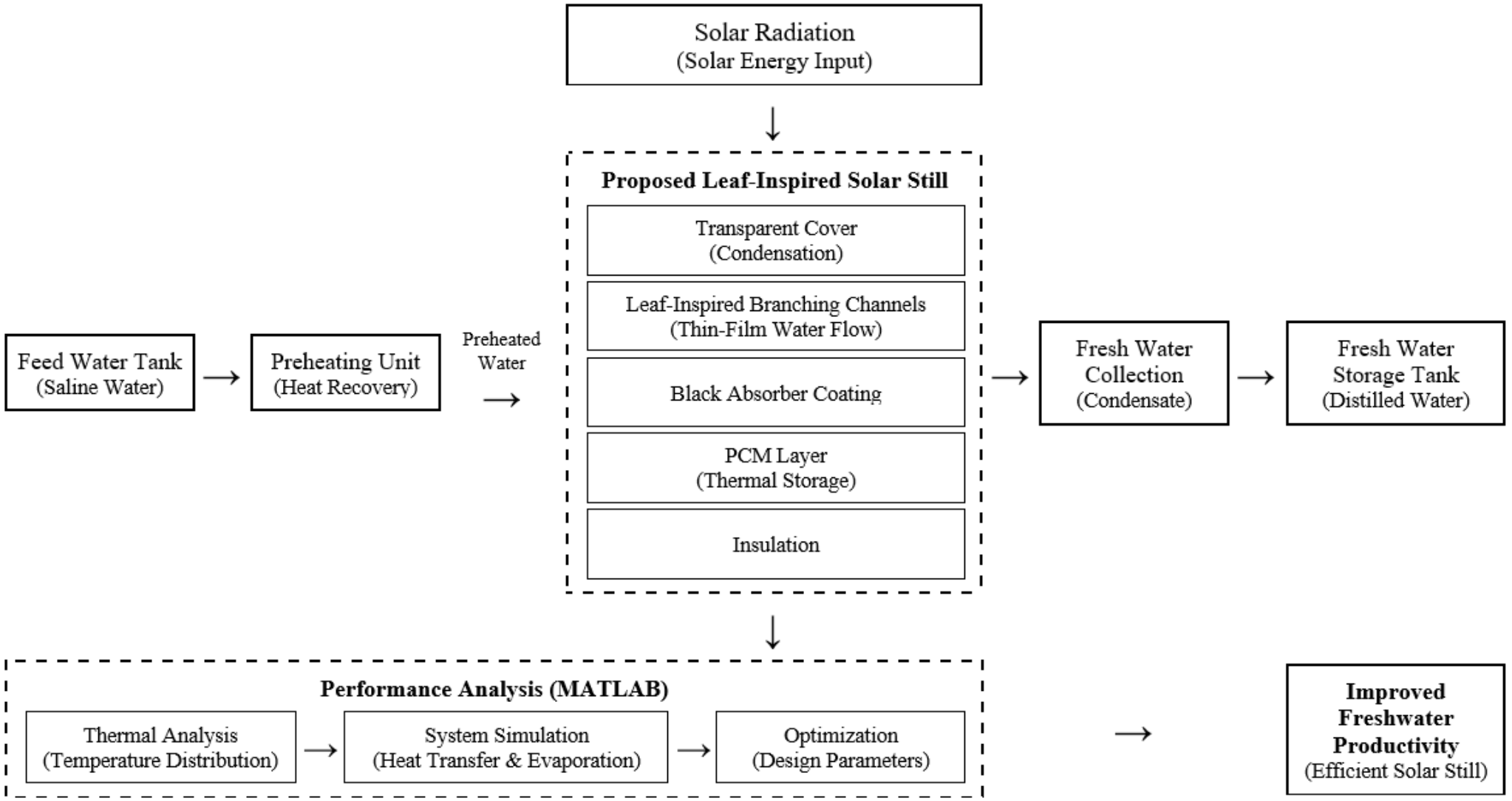
BLOCK DIAGRAM OF EXISTING SYSTEM



BLOCK DIAGRAM EXPLANATION

- Solar radiation enters the conventional solar still and provides heat energy.
- Saline or impure water from the feed water tank is supplied to the basin.
- The black-coated basin absorbs solar radiation and heats the water.
- The heated water evaporates and changes into water vapour.
- The water vapour condenses on the transparent glass cover.
- The condensed water flows into the freshwater collection unit.
- The collected freshwater is transferred to the freshwater storage tank.
- Thermal insulation reduces heat loss through the bottom and side walls.
- Some heat is lost to the surroundings, which reduces the system’s performance.
- The collected water is obtained through the solar distillation process.

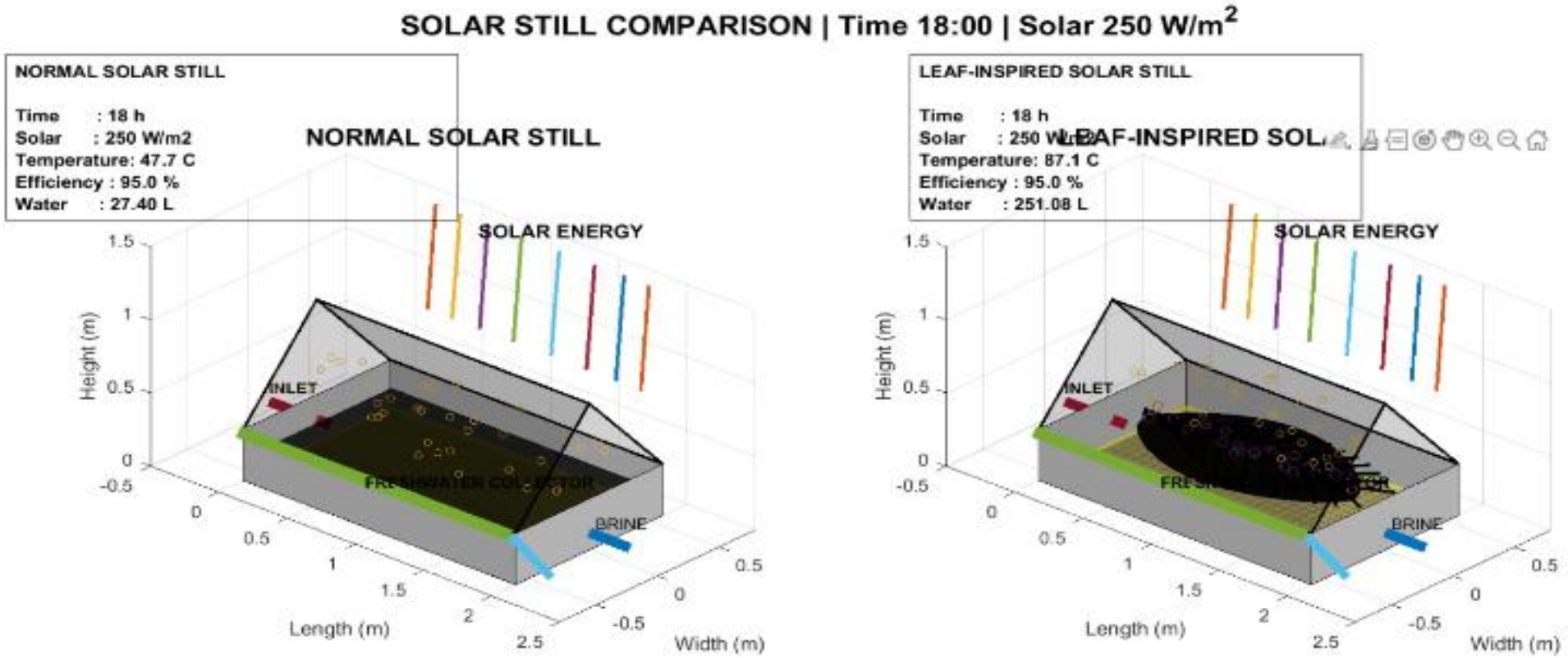
PROPOSED SYSTEM



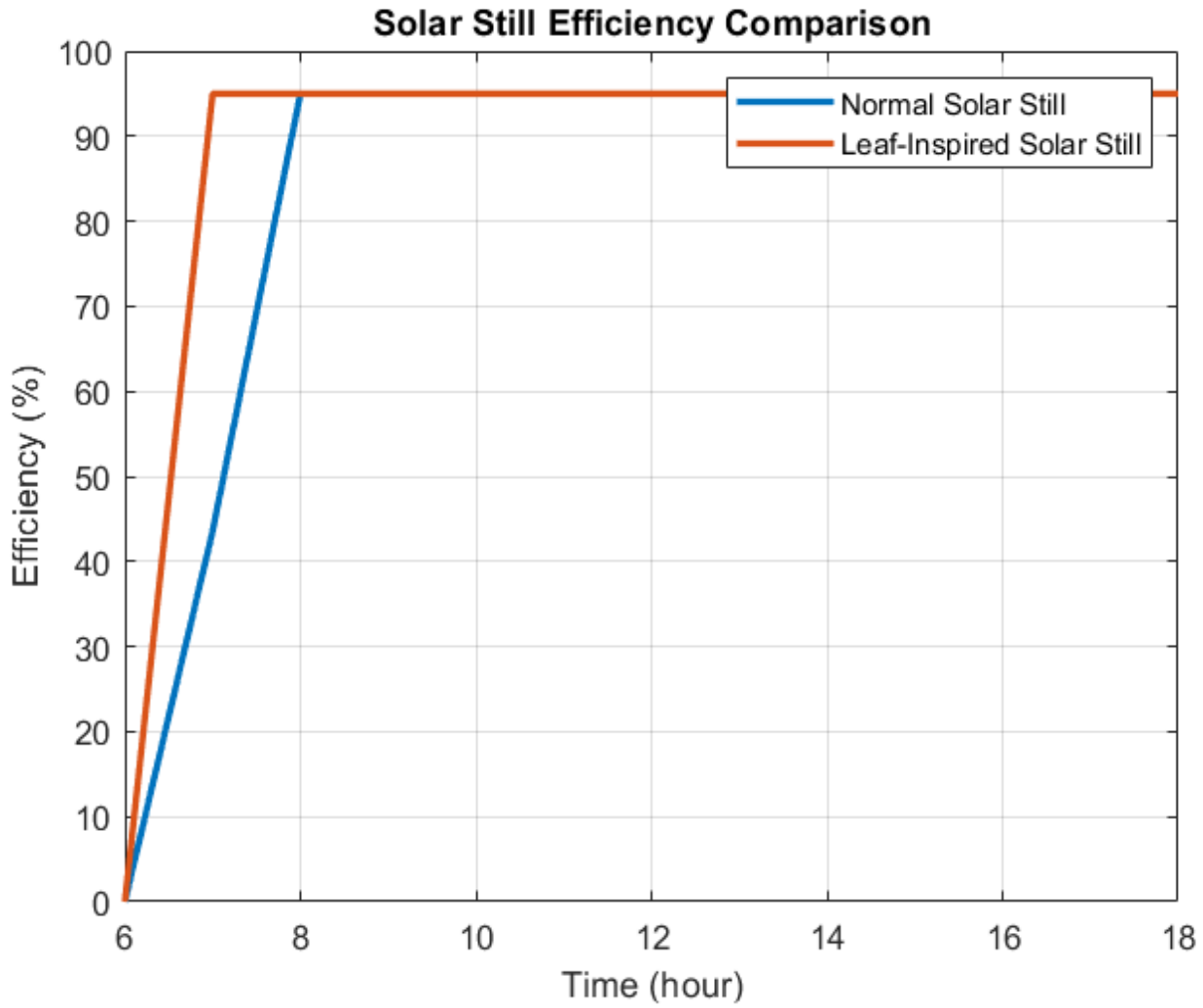
PROPOSED SYSTEM EXPLANATION

- Solar radiation enters the solar still and provides the required heat energy.
- Saline water from the feed water tank passes through the preheating unit.
- The preheating unit uses recovered heat to increase the water temperature.
- Preheated water flows through the leaf-inspired branching channels as a thin film.
- The black absorber coating absorbs solar energy and heats the water.
- The PCM layer stores excess heat and releases it when solar radiation decreases.
- The heated water evaporates, and the water vapour condenses on the transparent cover.
- The condensed freshwater flows into the collection unit and is stored in the freshwater tank.
- MATLAB is used to analyse temperature distribution, heat transfer, evaporation, and system performance.
- The design parameters are optimized to improve freshwater productivity and overall efficiency.

SIMULATION OUTPUT



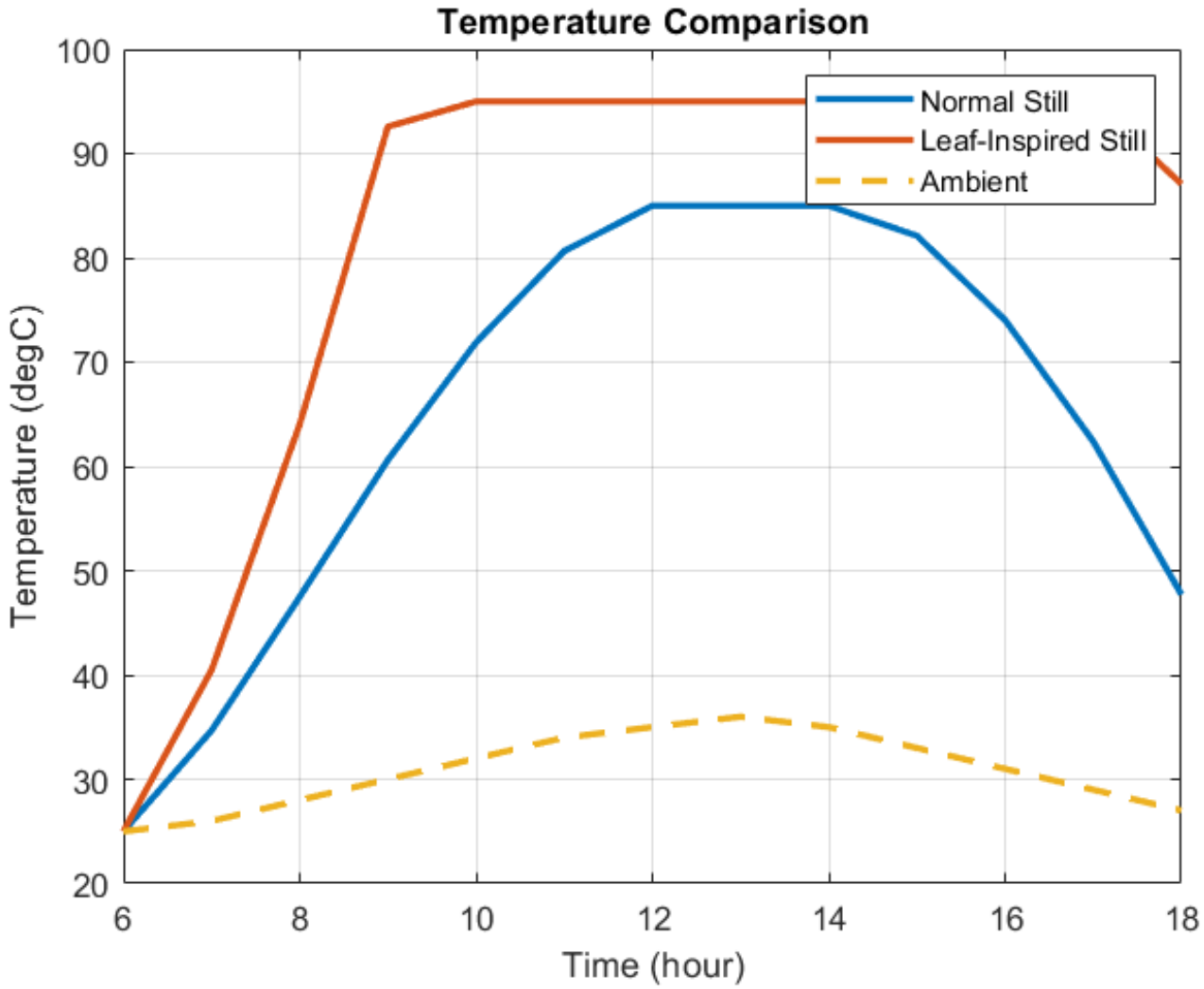
SOLAR STILL EFFICIENCY COMPARISON



EXPLANATION

- The efficiency of both systems increases during the morning.
- The normal solar still has an average efficiency of 90.71%.
- The leaf-inspired solar still reaches an average efficiency of 95.00%.
- The proposed system shows better efficiency than the normal solar still

TEMPERATURE COMPARISON



EXPLANATION

- The temperature of both solar stills increases during the daytime.
- The normal solar still reaches a maximum temperature of 85°C.
- The leaf-inspired solar still reaches a maximum temperature of 95°C.
- The proposed system shows a 10°C increase in maximum temperature.

CONCLUSION

The leaf-inspired solar still is designed to produce freshwater using solar energy. Its leaf-like structure helps distribute water evenly and improves the heating process. The system uses solar heat to evaporate water, which is then condensed and collected as freshwater. This design aims to improve the performance of a conventional solar still. It is a simple and eco-friendly method for freshwater production. Thus, the proposed system supports sustainable use of solar energy.

REFERENCE

1.D. Dsilva Winfred Rufuss, S. Iniyan, L. Suganthi and P. A. Davies, "Solar Stills: A Comprehensive Review of Designs, Performance and Material Advances," *Renewable and Sustainable Energy Reviews*, 2016.

2.A. A. M. Omara, A. A. A. Abuelnuor, H. A. Mohammed and M. Khiadani, "Phase Change Materials (PCMs) for Improving Solar Still Productivity: A Review," *Journal of Thermal Analysis and Calorimetry*, 2020.

3.V. P. Katekar and S. S. Deshmukh, "A Review of the Use of Phase Change Materials on Performance of Solar Stills," *Journal of Energy Storage*, 2020.

THANK YOU